Research

• 37 deaths – Coal-mining fatalities in Jharkhand during 2023–2025.

• 2023–2025 – Period covered by the mining fatality data.

• 2 major accident causes – Transportation/machinery accidents and ground movement incidents.

• 75% – Knowledge retention reported with VR/AR-based training.

• 10% – Knowledge retention reported with traditional training.

• 7.5× – Improvement in knowledge retention with VR/AR training.

• 45% – Reported reduction in workplace accidents using VR/AR training.

• 4× – Faster training compared with traditional training methods.

• 2 training approaches – Traditional classroom-based training and immersive VR/AR-based training.

Article :- https://link.springer.com/article/10.1007/s40789-025-00822-0

• **DGMS:** Directorate General of Mines Safety (DGMS), headquartered in **Dhanbad, Jharkhand**, is the main regulatory authority for mine safety in India.

• **Initial Training:** Workers must receive **vocational training before deployment**, covering essential mine-safety practices.

• **Refresher Training: Annual refresher training** is required to maintain workers' safety knowledge and skills.

• **Practical Training:** Training includes both **theoretical and practical instruction** through Vocational Training Centres (VTCs) and Group Vocational Training Centres (GVTCs).

• **Specialized Safety Training:** Additional training is required for **new equipment, hazardous operations, and specific safety responsibilities**.

Article :- OSH Code 2020.pdf

Training Delivery Methods:

- Classroom-based theoretical instruction
- Gallery training (physical mock-ups)
- On-the-job practical training under supervision
- Safety committees and workmen's inspectors provide ongoing safety awareness

Training Challenges:

- Safety training often treated as "checkbox compliance" rather than skill-building, contributing to 48,000+ factory accidents annually (38 deaths/day) across India.halldale+1

  Link :- https://www.halldale.com/safety-critical-industries/indias-factories-struggle-to-meet-safety-standards

- **Assumption:** Actual on-the-ground training quality varies widely between organised-sector mines (BCCL, CCL, Tata Steel) and unorganised/small-scale operations. The Dumka example suggests unorganised sector miners receive episodic awareness sessions rather than structured vocational training.
- **Assumption:** Classroom-based training remains the dominant mode; AR/VR is not yet standard practice, though pilot projects exist

Cost Constraints:

- SMEs (50-200 employees) face Rs 8-15 lakh implementation costs for AR/VR training, with per-employee costs of Rs 4,000-7,500.

  https://www.cartoonmango.com/insights/ar-vr-industrial-training-guide

Contextual Challenges Specific to Jharkhand

1. Language Diversity:

- Workforce includes Hindi, Santhali, Mundari, Khortha, and other tribal language speakers
- English/Hindi training materials fail to reach Santhali-dominant workers in mining belts (Dhanbad, Bokaro, Hazaribagh)

Rank	Language	Speakers	Percentage
1	**Khortha/Khotta**	77,38,960	**23.46%**
2	**Hindi**	70,59,131	**21.40%**
3	**Bengali**	32,12,831	**9.74%**
4	**Santali**	28,95,764	**8.78%**
5	**Urdu**	19,65,438	**5.96%**

India - C-16: Population by mother tongue, Jharkhand - 2011

3. Common Mining/Manufacturing Accidents and Their Causes

Fatal Accident Statistics:

- **Jharkhand leads India** in coal mine fatalities: 37 deaths (2023-2025), 12 deaths in 2023 alone, 22 deaths in 2025.etvbharat
- Out of 465 analyzed mining accident deaths, Jharkhand accounted for 100+ fatalities, with transportation machinery (non-winding) as predominant accident class.

Assumptions

- **Assumption:** Jharkhand's mining accident figures likely include a higher proportion of informal/illegal mining incidents that are underreported compared to DGMS data, which primarily captures regulated mines.
- **Assumption:** The transition to mechanisation-related accidents in opencast mines suggests a growing need for training on heavy equipment operation and maintenance safety.

Existing AR/VR Safety Training Solutions in India and Globally

india: Tata Power Training on Wheels, Cusmat, Associative
Global: Studio NORCAT (FiAR 2.0), Pixaera, Ludus

- **Immersafety** (India) – AR/VR safety modules for work-at-height, confined space, electrical safety, fire safety, etc.simbott

- **Simulanis** (India) – Custom AR/VR/XR safety simulators for enterprises like Cairn, including fire safety and SOP-to-XR training.simulanis

- **Senar** (by Immersive Factory, EU/Global) – Mobile AR safety training that runs on phones/tablets with ready-to-use occupational risk simulators.

- **STRIVR** (US/Global) – Enterprise VR/AR safety training platform with authoring, tracking, and analytics, widely used in high-risk industries.

- **Immersive Factory** (EU/Global) – AR training catalog and 30+ immersive workshops for heavy industry clients (e.g., TotalEnergies, Airbus).

- **TeamViewer Frontline** (Germany/Global) – Enterprise AR platform for connected workers, including safety procedures, training, and guided work.

PTC Vuforia (US)

- Enterprise AR platform with strong manufacturing focus
- Pricing: $50K-$300K (Rs 40L-2.5Cr); English-only; remote support only.

Key Limitations Across Existing Solutions (Verified from Literature)

A 2026 literature review on VR/AR safety training in high-risk industries identified these implementation challenges:

1. **High costs** of content development and hardware.
2. **Need for specialised instructor training** — existing VTC instructors may lack XR competency.
3. **Scenario localisation requirements** — generic simulations do not reflect local mine conditions, language, or regulatory context.
4. **Limited viewing fields, motion sickness, and control issues** in XR headsets, stemming from human and socio-technical factors.
5. **Lack of long-term impact evidence** — studies show short-term gains but no evidence for sustained behavioural change.
6. **Device compatibility and high implementation costs** hinder widespread adoption.
7. **Assumption:** None of the existing competitors have specifically adapted their content for Jharkhand's linguistic diversity (Hindi, Santhali, Kurukh, Mundari, Bengali, Odia) or its specific mix of opencast and underground mines.

8. **Assumption:** Existing solutions are priced for large enterprises; Jharkhand's SME manufacturing sector and unorganised mines cannot afford them.
9. **Assumption:** Most solutions are VR-first rather than AR-first; AR-on-mobile may be more scalable for Jharkhand's workforce and infrastructure constraints.

Opportunities That Existing Solutions Are Missing

Based on the research, here are the key gaps and opportunities for your AR-based vocational training simulator:

Opportunity 1: Mobile-First AR for the Unorganised Sector

Gap: Existing solutions (CHRP-INDIA, Cusmat) are VR-heavy, requiring expensive headsets and controlled environments. Deployment has focused on large PSU mines.
Opportunity: Build an **AR-on-smartphone** solution that works on mid-range Android devices. This addresses the unorganised sector (as highlighted by the Dumka unorganised miners' camp) and small-scale manufacturing units that cannot afford VR infrastructure. Smartphones are already ubiquitous in India, and ARCore/ARKit enable marker-based and markerless AR without expensive hardware.

Opportunity 2: Multi-Lingual, Voice-First Design

Gap: No existing solution addresses Jharkhand's linguistic diversity (Hindi, Santhali, Kurukh, Mundari, Bengali, Odia) or the low literacy levels among workers (67% studied only up to Class 10 or below).
Opportunity: Design a **voice-guided, icon-driven AR interface** with support for at least Hindi, Santhali, and Kurukh. Safety instructions should be visual and demonstrative, not text-heavy.

Opportunity 3: DGMS/OSH Code Compliance Tracking and Certification

Gap: DGMS has standardised Forms T(2) for training certificates, and Rule 164 mandates "computers, simulators" for VTCs. But no existing AR training tool provides **automated compliance tracking, assessment scoring, and certificate generation** aligned with OSH Code 2020 requirements.
Opportunity: Build a **DGMS-aligned assessment engine** that automatically generates

digital training records, tracks refresher training due dates, and produces evidence for DGMS inspections. This directly addresses the High Court's concerns about inadequate safety compliance in Jharkhand mines.

Opportunity 4: Digital Twin of Jharkhand-Specific Mine Conditions

Gap: Most VR/AR content is generic or adapted from global templates. The HoloXperience platform uses digital twin technology but is IIT ISM's proprietary project, not a widely deployable training tool.
Opportunity: Create a **library of AR training modules specific to Jharkhand's mine types** — opencast coal mines (like Lal Matia, Hazaribagh), underground coal mines (BCCL areas), iron ore mines (Noamundi, Joda), and copper mines (Ghatsila). Include scenarios for roof fall, gas leak, explosives handling, heavy machinery operation, and furnace safety in steel plants.

Opportunity 5: Low-Cost, Cloud-Based Content Authoring

Gap: Literature consistently identifies **high content development cost** as a primary barrier to XR adoption.
Opportunity: Develop a **cloud-based authoring tool** that allows DGMS trainers and VTC instructors to create or modify AR training scenarios without coding. This addresses the instructor training gap and reduces content localisation costs.

Opportunity 6: Blended Learning with On-Site Assessment

Gap: Research shows VR/AR training produces strong short-term gains but **no evidence for long-term impact**. Existing solutions are often one-time or periodic interventions.
Opportunity: Design a **blended learning system** where AR training is followed by on-site competency assessment using the same AR app (e.g., a worker is asked to identify hazards in their actual work environment, with the app scoring their responses). This creates a continuous learning loop aligned with the **annual refresher training** requirement under MVTR.

Opportunity 7: Integration with Jharkhand's Centralised Monitoring System

Gap: Jharkhand is planning centralised online monitoring of industries for labour and safety compliance, but no AR training tool feeds data into such a system.
Opportunity: Design the solution with **APIs for integration** with the Jharkhand Labour Department's monitoring platform, allowing training completion data, competency scores, and compliance status to flow directly to regulators. This could significantly enhance the state's enforcement capability.

Opportunity 8: Focus on Steel and Manufacturing Plants in Jharkhand

Gap: Most AR/VR safety solutions focus on mining. Jharkhand's manufacturing sector (steel plants in Jamshedpur, Ramgarh, Bokaro; auto component units) faces distinct hazards — furnace explosions, hot metal contact, crush injuries from power presses.
Opportunity: Develop **manufacturing-specific AR modules** for induction furnace safety, hot metal handling, machine guarding, lockout/tagout procedures, and PPE compliance. This is an underserved segment.

Opportunity 9: Trainer-in-the-Loop Design

Gap: Rule 164 requires qualified training officers, but there is a shortage of instructors with XR competency.
Opportunity: Build the AR tool as a **trainer augmentation tool** rather than a trainer replacement — it provides the simulation, while the human trainer provides context, debriefing, and behavioural coaching. This reduces the training burden on instructors while keeping them central to the learning process.

1. **Language Inclusion:** Hindi + Santhali + English trilingual support (voice + text)
2. **Hardware Agnostic:** Quest 3 VR + smartphone AR (WebXR) options
3. **Offline-First:** Zero-dependency on internet for core training
4. **NSQF-Aligned:** Certificates mapped to NOS codes for formal recognition
5. **DGMS-Compliant:** Training records aligned with accident classification codes
6. **MSME-Affordable:** Modular pricing (Rs 50K/module) for small enterprises
7. **Jharkhand-Specific Content:** Coal + metalliferous mining + steel/manufacturing scenarios
8. **Evidence-Based:** Modules designed around top accident causes (transportation, ground movement, explosives)link.springer+1

Verified Facts vs Assumptions

Verified Facts

- Jharkhand leads India in mining fatalities (37 deaths 2023-2025)etvbharat
- 31% of coal mine fatalities from transportation machinery, 24% from falls, 16% from ground movementdgms
- VR training improves knowledge retention to 75% vs 10% for lecturescartoonmango
- MVT Rules 1966 mandate pre-employment training for all mine workersscribd+1
- IIT (ISM) Dhanbad launched India's first VR mine simulator (Nov 2025)pib.gov+1
- 60% of factory workers prefer regional language trainingcartoonmango+1
- 84% of organizations lack AI-enabled training implementation knowledgeleap10x

Assumptions (Require Validation)

- Santhali/Mundari language support will significantly improve training effectiveness (no studies exist)
- Smartphone-based AR can replace headset VR for complex equipment training
- MSMEs will adopt AR/VR training if priced below Rs 5 lakh
- Offline-first design will not compromise learning analytics quality
- NSQF-aligned certificates will incentivize worker participation more than employer mandates

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Recommendations for Solution Design

1. **Prioritize Transportation Machinery + Ground Movement Modules** (47% of fatalities combined)dgms
2. **Design for Meta Quest 3 + Android Smartphones** (cost-effective hardware stack)cartoonmango
3. **Implement Hindi + Santhali Voice-Over Minimum** (覆盖 95%+ workforce)cartoonmango+1
4. **Offline-First Architecture with Delayed Sync** (accommodate connectivity gaps)cartoonmango
5. **NSQF NOS Code Mapping** (MIN/N1211, MIN/N1212, MIN/N1702-04) for certificationnqr.gov
6. **DGMS Accident Code Analytics** (track training impact on Code 335, 111, etc.)link.springer+1
7. **15-Minute Micro-Learning Modules** (shift-compatible deployment)leap10x
8. **Pilot with IIT (ISM) Dhanbad + Coal India Subsidiaries** (leverage existing VRMS partnership)pib.gov

This research foundation provides evidence-based direction for designing a contextually appropriate, impactful AR-based vocational training simulator for Jharkhand's mining and manufacturing sectors.

FEASIBILITY AND VIABILITY

A. Technical Feasibility

- **Android-first AR:** Unity + AR Foundation + ARCore; low-poly assets for affordable devices.
- **Offline-first:** Scenarios, scores and language packs work without continuous internet.
- **SIH-ready MVP:** 2 environments, 5 hazards, 3 languages, quiz, scoring and instructor dashboard.
- **Future-ready:** AI PPE vision, speech interaction, digital twins and enterprise analytics.

B. Evidence / Need

- India coal mines recorded **38 fatal accidents and 49 fatalities in 2024**.dgms
- Leading causes: **transportation 31%, falls 24%, machinery 18%, ground movement 16%**.dgms
- Jharkhand recorded **12 coal-mine fatalities in 2023**; 2024 figure was **3 provisional**.sansad
- VR meta-analysis: effectiveness over traditional training of **0.593 behaviour, 0.432 skills and 0.777 experience**.pubmed.ncbi.nlm.nih

C. Challenges & Mitigation

- **Connectivity/device gap** → offline content, shared tablets and non-AR fallback.
- **Language/usability** → Hindi/English/local audio, icons and short sessions.
- **AI/SOP accuracy** → rule-based MVP, expert validation and human-approved content.
- **Adoption/privacy** → instructor-led rollout, minimum data and role-based dashboards.

D. Viability & Scalability

- Estimated MVP: **₹6.5–13 lakh**; pilot: **₹20–40 lakh**.
- Paying customers: **mines, factories, ITIs, government and CSR programmes**.
- Revenue: institutional licence + annual maintenance + customised modules.
- Roadmap: **Jharkhand mining → Indian mining → manufacturing → construction/logistics**.

SECTION 1 — POTENTIAL IMPACT ON THE TARGET AUDIENCE

1. **Mining workers:** Practice underground evacuation, roof-fall, gas-leak, fire and rescue procedures through AR overlays in realistic mine-like scenarios—without entering dangerous zones.

2. **Manufacturing workers:** Repeatedly train on lockout/tagout, confined-space entry, electrical safety, machine guarding and fire response, improving hazard recognition and retention of safe procedures.

3. **New recruits:** Get risk-free pre-deployment onboarding where they can make emergency decisions, learn PPE/exit/alarm protocols and build confidence before working on live mine or plant floors.

4. **ITI/polytechnic/vocational students:** Use industry-relevant AR safety modules that connect classroom theory to actual mining and manufacturing hazards, making them more job-ready for Jharkhand industries.

5. **Safety trainers:** Deliver standardized, repeatable AR lessons and assessments across batches and sites, reducing dependence on live mock drills and manual demonstrations.

SECTION 2 — BENEFITS OF THE SOLUTION

SOCIAL BENEFITS

6. Builds a stronger safety culture by normalizing repeated hazard-recognition and emergency-response practice among workers, trainees and trainers.

7. Reduces unsafe practices because learners can make mistakes in AR, see consequences and correct their actions before facing real workplace risk.

ECONOMIC BENEFITS

8. Lowers training costs by supplementing expensive physical mock drills, external trainers and travel with reusable smartphone/tablet-based AR modules.

9. Reduces operational downtime because workers can train during breaks or off-shift without stopping production or blocking active mine/plant areas.

OPERATIONAL / TECHNICAL BENEFITS

10. Smartphone-based AR keeps training low-cost and scalable, while reusable modular scenarios allow easy expansion to new hazards, equipment and industrial layouts.

11. Built-in performance tracking and assessment captures module completion, hazard identification, emergency choices and repeat attempts for trainer review.

ENVIRONMENTAL BENEFITS

12. Reduces repeated physical training setups and consumables, lowering resource consumption and waste from mock drills and temporary training infrastructure.

SECTION 1 — POTENTIAL IMPACT ON THE TARGET AUDIENCE

- **Mining workers:** Practise hazard identification, PPE use, evacuation and emergency response in realistic mine-related scenarios without exposure to physical danger.
- **Manufacturing workers:** Develop safer responses to machinery hazards, fire incidents, chemical exposure and workplace accidents through guided AR simulations.
- **New recruits:** Gain practical safety awareness and confidence before entering mining or manufacturing work environments.
- **ITI, polytechnic and vocational students:** Bridge the gap between classroom instruction and industrial safety practice through interactive, location-independent training.
- **Safety trainers:** Deliver standardized demonstrations, repeat difficult scenarios, and assess trainee performance using the same safety procedures across batches.
- **All trainees:** Improve retention and emergency decision-making through repeated practice, immediate feedback and risk-free learning.

SECTION 2 — BENEFITS OF THE SOLUTION

SOCIAL BENEFITS

- **Stronger safety culture:** Makes safe behaviour, hazard recognition and emergency preparedness a regular part of workforce training.
- **Reduced unsafe practices:** Reinforces correct responses to industrial hazards before workers encounter them during actual operations.

ECONOMIC BENEFITS

- **Lower training costs:** Uses smartphones and reusable digital scenarios, reducing dependence on costly physical mock-drill arrangements.
- **Less operational disruption:** Enables training without repeatedly stopping machinery, occupying production areas or deploying full-scale emergency setups.

OPERATIONAL / TECHNICAL BENEFITS

- **Affordable and scalable delivery:** Smartphone-based AR can extend standardized training to mines, factories, ITIs and remote training centres.
- **Measurable skill development:** Records task completion, response accuracy and training progress for structured assessment and targeted retraining.
- **Reusable hazard scenarios:** Allows repeated practice and quick expansion to hazards such as fire, machinery accidents, chemical exposure, electrical faults and mine emergencies.

ENVIRONMENTAL BENEFITS

- **Lower training resource consumption:** Reduces repeated use of physical materials, fuel, equipment and space for demonstration-based safety drills.